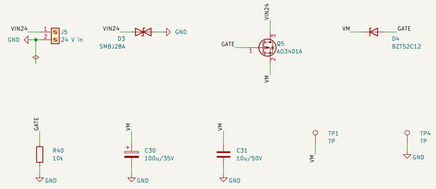
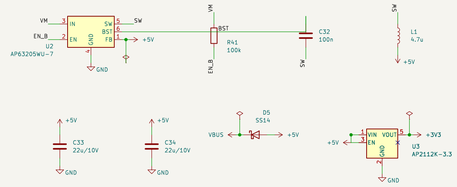
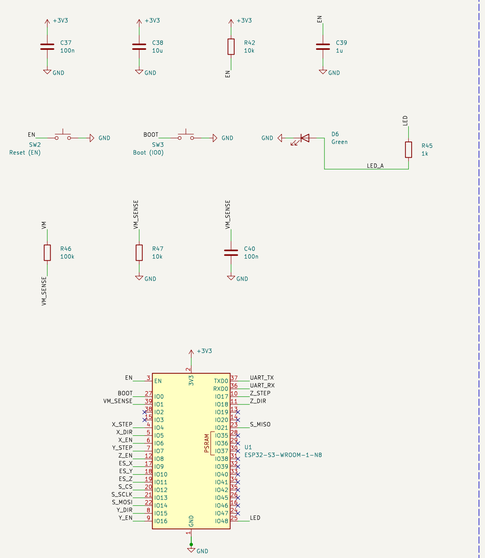
POWER IN: 24 V, TVS, reverse-polarity PMOS with 12 V gate clamp, bulk



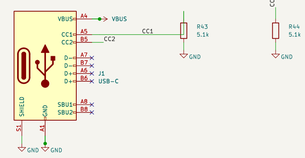
5 V BUCK AP63205 + 3.3 V LDO



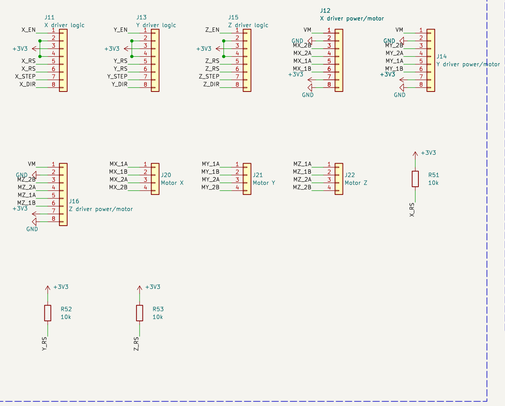
CONTROLLER: ESP32-S3 (native USB, step/dir for 3 axes, SPI sensor)



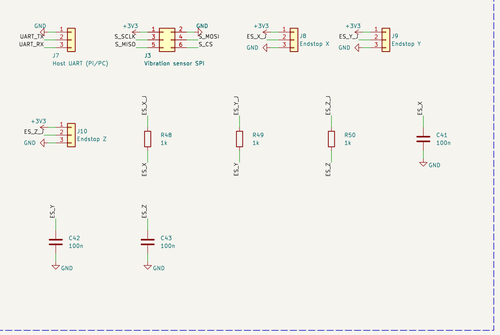
USB-C



DRIVER SOCKETS (A4988 / DRV8825 / TMC2209 StepStick) + MOTOR OUTPUTS



ENDSTOPS, HOST UART, VIBRATION SENSOR



KINETIC: 24 V in (12–28 V), 5 V buck, 3.3 V LDO. ESP32-S3 runs the jerk-limited motion planner (kn_motion) and the vibration classifier (kn_vib). Three StepStick sockets; microstep pins MS1–3 are tied to 3.3 V (1/16 on A4988, 1/32 on DRV8825) and RESET/SLEEP are pulled up. Current limit is set on each driver module. Endstops are normally-closed to GND recommended.

First engineering release; see pcb/README.md for status and limits

ClaudeWork

Sheet: /

File: kinetic_ctl.kicad_sch

Title: KINETIC stepper / servo motion controller — main board

Size: A1	Date: 2026-11-01
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	**Rev: A**
	Id: 1/1